Eine Woche, ein Beispiel

10.20 Schur functor: basic formulas

Main reference:
[FH]: Willian Fulton and Joe Harris. Representation Theory. A First Course.

[Hall]: Brian Hall. Lie Groups, Lie Algebras, and Representations: An Elementary Introduction. Graduate Texts in Mathematics. Cham: Springer International Publishing, 2015.
https://ocw.mit.edu/courses/18-755-lie-groups-and-lie-algebras-ii-spring-2024/resources/mit18_755_s24_lec02_pdf/

Álvaro Gutiérrez Cáceres, The combinatorics of symmetric polynomials in types A, B, and C
https://www.math.uni-bonn.de/ag/stroppel/Gutierrez%20-%20The%20combinatorics%20of%20symetric%20polynomials%20in%20types%20A,%20B,%20and%20C-1.pdf

In this document, $\operatorname{char} k = 0$, $V \in \mathrm{Vect}_k$.

Schur fctor helps us to decompose $V^{\otimes k}$ by S_k gp action.
$\mathbb{S}^\lambda V$ generalize $\mathrm{Sym}^k V$ & $\Lambda^k V$. Moreover,

$$\begin{array}{ccc} \mathrm{Rep}(GL(V)) & = & \mathrm{Rep}(A_{n-1}) \\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ \mathbb{S}^\lambda V & = & \mathcal{L}(\lambda) \end{array} \qquad n = \dim V$$

Here, λ has many expressions,
e.g.

	partitions	weights
$\lambda =$	$= (3,1)$	$= 2\varpi_1 + \varpi_2$
$\lambda =$	$\cdots = (\lambda_1, \lambda_2, \cdots)$	$= \sum m_i \varpi_i$

$\{h_{ij}\} =$ hook length

1. dimension
2. $\mathbb{S}^\lambda(V \oplus W)$ and ...

1. dimension

It can be computed from the Weyl dimension formula:

$$\dim_{\mathbb{C}} L(\lambda) = \frac{\prod_{\alpha\in\Delta^+} (\lambda+\rho, \alpha)}{\prod_{\alpha\in\Delta^+} (\rho, \alpha)}$$

$$\overset{\substack{L(\lambda)\in \mathrm{rep}(A_{n-1}) \\ \lambda = \sum m_i \varpi_i}}{=} \frac{(m_1-1)\cdots(m_{n-1}+1)\,(m_1+m_2+2)(m_2+m_3+2)\cdots \quad \cdots \quad (\sum m_i + n-1)}{\underbrace{1 \cdots 1}_{n-1 \text{ many}} \quad \underbrace{2 \cdot 2 \cdots}_{n-2 \text{ many}} \cdots \quad n-1}$$

$$= \prod_{1\le i<j\le n} \frac{\lambda_i - \lambda_j + j - i}{j-i} \qquad \text{[FH, Thm 6.3 (1)]}$$

$$= \prod_{1\le i<j\le n} \frac{n-i+j}{h_{ij}} \qquad \text{[FH, Ex 6.4]}$$

Following [Hall Example 10.23],

$$\Delta^+ = \left\{ \begin{array}{ccccc} & & \sum\alpha_i & & \\ & \iddots & & \ddots & \\ \alpha_1+\alpha_2 & & \cdots & & \alpha_{n-2}+\alpha_{n-1} \\ \alpha_1 & \alpha_2 & \cdots & & \alpha_{n-1} \end{array} \right\}$$

$$\rho = \frac{1}{2} \sum_{\alpha\in\Delta^+} \alpha$$

$$= \frac{1}{2} \sum_{i=1}^{n-1} i(n-i)\alpha_i$$

$$= \sum_{i=1}^{n-1} \varpi_i \qquad\qquad \chi = \sum_{i=1}^{n-1} \langle \chi, \alpha_i \rangle \varpi_i$$

These would be enough to explain the first equality above.

E.g. When $\lambda = (3,1) = \begin{array}{|c|c|c|}\hline & & \\ \hline & & \\ \hline \end{array}\!\!\begin{array}{|c|}\hline \\ \hline \\ \hline \end{array} = 2\varpi_1 + \varpi_2$,

$$\dim_k \mathcal{L}(\lambda) = \frac{(2+1)(1+1)(0+1)\cdots \quad (2+1+2)(1+0+2)(0+0+2)\cdots \quad (2+1+(n-1))}{1\cdot 1\cdot 1\cdots \quad 2\cdot 2\cdot 2\cdots \quad n-1}$$

$$= \frac{(2+1)(1+1) \quad (2+1+2)(1+0+2) \quad \cdots (2+1+(n-1))}{1\cdot 1 \quad 2\cdot 2 \quad \cdots \quad n-1}$$

$$= \left(\frac{2+1}{1}\cdot\frac{3+2}{2}\cdot\frac{3+3}{3}\,0\cdots 00\,\frac{3+n-1}{n-1}\right)\cdot\left(\frac{1+1}{1}\cdot\frac{1+2}{2}\cdot\frac{1+3}{3}\cdots\frac{1+n-2}{n-2}\right)$$

$$= \frac{(n-1)\,n\,(n+1)(n+2)}{1\cdot 4\cdot 2\cdot 1}$$

$$= \frac{\begin{array}{|c|c|c|}\hline n & n+1 & n+2 \\ \hline n-1 & & \\ \hline \end{array}}{\begin{array}{|c|c|c|}\hline 4 & 2 & 1 \\ \hline 1 & & \\ \hline \end{array}}$$

2. $\mathbb{S}^{\lambda}(V \oplus W)$ and ...

E.g. $$(V \oplus W)^{\otimes 2} = V^{\otimes 2} \oplus (V \otimes W)^{\oplus 2} \oplus W^{\otimes 2}$$

$$\mathrm{Sym}^2(V \oplus W) = \mathrm{Sym}^2 V \oplus V \otimes W \oplus \mathrm{Sym}^2 W$$
$$\Lambda^2(V \oplus W) = \Lambda^2 V \oplus V \otimes W \oplus W^{\otimes 2}$$

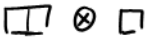
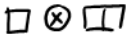
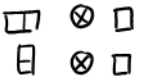
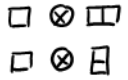
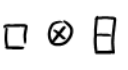
$$(V \oplus W)^{\otimes 3} = V^{\otimes 3} \oplus (V^{\otimes 2} \otimes W)^{\oplus 3} \oplus (V \otimes W^{\otimes 2})^{\oplus 3} \oplus W^{\otimes 3}$$

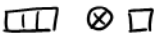
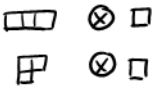
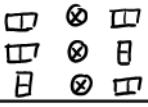
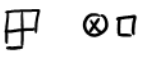
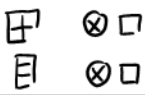
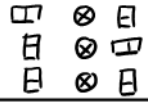
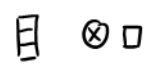
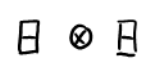
$$\mathrm{Sym}^3(V \oplus W) = \mathrm{Sym}^3 V \oplus \mathrm{Sym}^2 V \otimes W \oplus V \otimes \mathrm{Sym}^2 W \oplus \mathrm{Sym}^3 W$$
$$\mathbb{S}^{\ydiagram{2,1}}(V \oplus W) = \mathbb{S}^{\ydiagram{2,1}} V \oplus V^{\otimes 2} \otimes W \oplus V \otimes W^{\otimes 2} \oplus \mathbb{S}^{\ydiagram{2,1}} W$$
$$\Lambda^3(V \oplus W) = \Lambda^3 V \oplus \Lambda^2 V \otimes W \oplus V \otimes \Lambda^2 W \oplus \Lambda^3 W$$

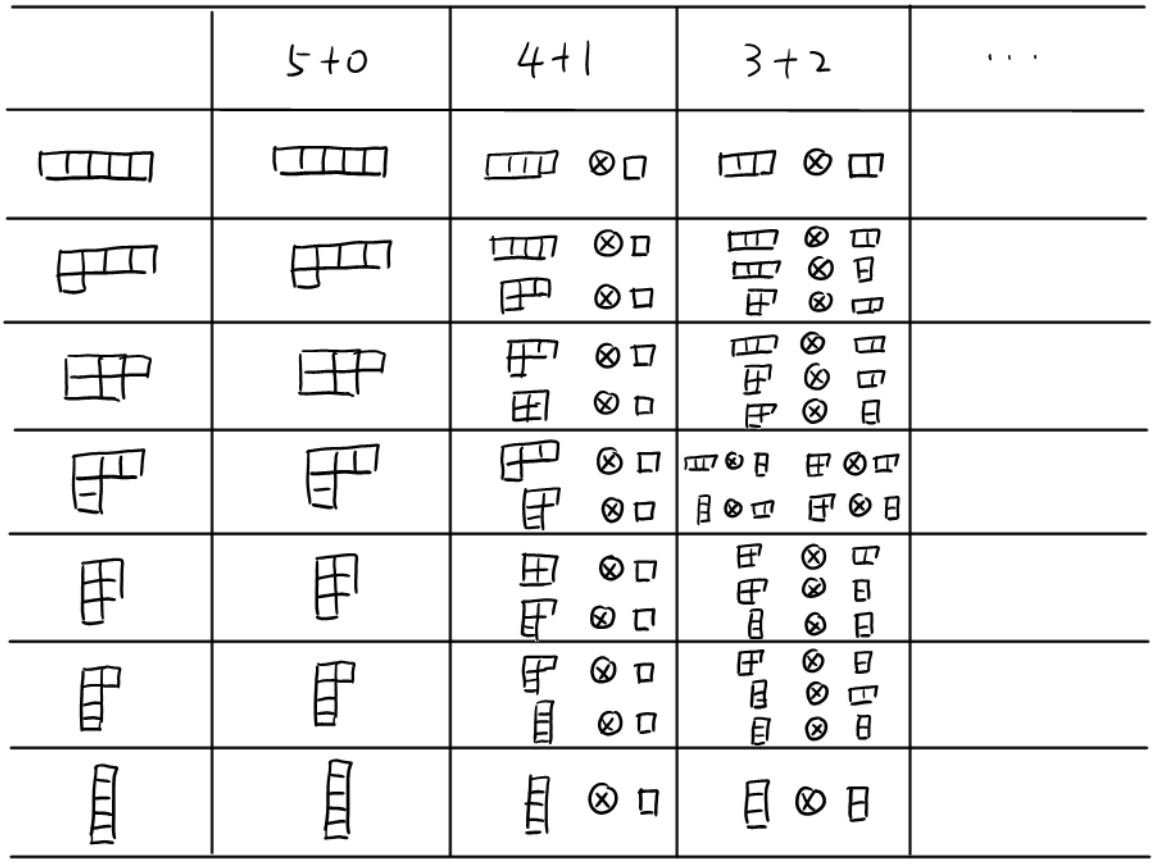
$$\frac{\begin{array}{|c|c|}\hline m+n & m+n+1 \\ \hline m+n-1 & \\ \cline{1-1}\end{array}}{\begin{array}{|c|c|}\hline 2 & 1 \\ \hline 1 & \\ \cline{1-1}\end{array}} = \frac{\begin{array}{|c|c|}\hline m & m+1 \\ \hline m-1 & \\ \cline{1-1}\end{array}}{\begin{array}{|c|c|}\hline 2 & 1 \\ \hline 1 & \\ \cline{1-1}\end{array}} + m^2 n + n m^2 + \frac{\begin{array}{|c|c|}\hline n & n+1 \\ \hline n-1 & \\ \cline{1-1}\end{array}}{\begin{array}{|c|c|}\hline 2 & 1 \\ \hline 1 & \\ \cline{1-1}\end{array}}$$

With the help of [FH,Ex 6.11], we can get the following tables:

	$2+0$	$1+1$	$0+2$
$\ydiagram{2}$	$\ydiagram{2}$	$\ydiagram{1} \otimes \ydiagram{1}$	$\ydiagram{1,1}$
$\ydiagram{1,1}$	$\ydiagram{1,1}$	$\ydiagram{1} \otimes \ydiagram{1}$	$\ydiagram{1,1}$

	3+0	2+1	1+2	0+3
				
				
				

	4+0	3+1	2+2	...
				
				
				
				
				



https://math.stackexchange.com/questions/84103/characters-of-symmetric-and-antisymmetric-powers

We have collected many remarkable coefficients from combinatorics, presented in the following formulas:

$N_{\lambda\mu\nu}$: Littlewood - Richardson number

$$\mathbb{S}^{\nu}(V \oplus W) \cong \bigoplus_{\mu,\lambda} N_{\lambda\mu\nu} \left(\mathbb{S}^{\lambda} V \otimes \mathbb{S}^{\mu} W\right) \qquad [\text{Ex } 6.11(a)]$$

$$\mathbb{S}^{\lambda} V \otimes \mathbb{S}^{\mu} V \cong \bigoplus_{\nu} N_{\lambda\mu\nu} \, \mathbb{S}^{\nu} V \qquad [\text{Ex } 6.19]$$

$$\mathbb{S}^{\nu/\lambda} V \cong \bigoplus_{\mu} N_{\lambda\mu\nu} \, \mathbb{S}^{\mu} V \qquad [\text{p}79, (6.7)]$$

As a Corollery,

$$\mathrm{Res}^{GL_{d_1+d_2}}_{GL_{d_1} \times GL_{d_2}} \mathbb{S}^{\nu}(\mathbb{C}^{d_1+d_2}) \cong \bigoplus_{\mu,\lambda} N_{\lambda\mu\nu} \, \mathbb{S}^{\mu}(\mathbb{C}^{d_1}) \otimes \mathbb{S}^{\lambda}(\mathbb{C}^{d_2})$$

$$\mathrm{Ind}^{S_{d_1+d_2}}_{S_{d_1} \times S_{d_2}} S^{\lambda} \otimes S^{\mu} \cong \bigoplus_{\nu} N_{\lambda\mu\nu} \, S^{\nu}$$

$$S^{\nu/\lambda} \cong \bigoplus_{\mu} N_{\lambda\mu\nu} \, S^{\mu}$$

$M_{\lambda\mu\nu}$: Plethysm coefficient

$$\mathbb{S}^{\mu}(\mathbb{S}^{\lambda} V) \cong \bigoplus_{\nu} M_{\lambda\mu\nu} \, \mathbb{S}^{\nu} V \qquad [\text{Ex } 6.17]$$

Later, $\lambda, \mu, \nu \vdash d$.

$g_{\lambda\mu\nu}$: Kronecker coefficient

$$\mathbb{S}^{\nu}(V \otimes W) \cong \bigoplus_{\mu,\lambda} g_{\lambda\mu\nu} \left(\mathbb{S}^{\lambda} V \otimes \mathbb{S}^{\mu} W\right) \qquad [\text{Ex } 6.11\,(b)]$$

$$S^{\lambda} \otimes S^{\mu} \cong \bigoplus_{\nu} g_{\lambda\mu\nu} S^{\nu} \qquad [\text{Ex } 4.51]$$

where

$$g_{\lambda\mu\nu} = \sum_{\xi \vdash d} \frac{1}{z_{\xi}} \chi_{\lambda}(C_{\xi}) \chi_{\mu}(C_{\xi}) \chi_{\nu}(C_{\xi}) \qquad [\text{Ex } 4.51]$$

$$z_{\xi} = \frac{d!}{|C_{\xi}|} = \prod_j (j^{\xi_j} \xi_j!) \qquad C_{\xi}: \text{conj class crspd to } \xi$$

$K_{\lambda\mu}$: Kostka number

https://mathoverflow.net/questions/314594/about-relation-between-kostka-numbers-and-littlewood-richardson-coefficient

$$\Lambda^{\mu_1} V \otimes \cdots \otimes \Lambda^{\mu_r} V \cong \bigoplus_{\lambda} K_{\lambda\mu} \mathbb{S}^{\lambda^T} V \qquad [\text{Ex } 6.13]$$

$$S_{\lambda} = \sum_{\mu} K_{\lambda\mu} M_{\mu} \qquad [(A.19)]$$

Left:

$$\mathbb{S}^{\lambda} V \otimes \mathbb{S}^{\mu} V \cong \ ? \qquad \text{inner tensor product}$$

$$\mathbb{S}^{\mu} \mathbb{S}^{\lambda} V \cong \ ? \qquad \text{inner plethysm}$$

The inner action is given by

$$GL_d \xrightarrow{\Delta} GL_d \times GL_d \subset GL_{2d} \circlearrowright V$$

the inner plethysm of s-symmetric functions
https://www.math.ucla.edu/~pak/hidden/papers/Thibon-Plethysm.pdf

Description of $\mathbb{S}_\lambda V$. [Ex 6.14, 6.19, 6.20]

$\mu = \lambda^T$, $\tilde{\mu} = \tilde{\lambda}^T$

$$\begin{aligned}
\mathbb{S}^\lambda V &= \mathrm{Im}\,(c_\lambda|_{V^{\otimes d}}) \\
&= \mathrm{Im}\,\Big(\bigotimes_i \Lambda^{\mu_i} V \longrightarrow \otimes V^{\otimes d} \longrightarrow \bigotimes_j \mathrm{Sym}^{\lambda_j} V \Big) \\
&= \mathrm{Im}\,\Big(\bigotimes_j \mathrm{Sym}^{\lambda_j} V \longrightarrow \otimes V^{\otimes d} \longrightarrow \bigotimes_i \Lambda^{\mu_i} V \Big) \\
&= \mathrm{Ker}\,\big(\mathbb{S}^{(\lambda_1, \dots, \lambda_{k-1})} V \otimes \mathrm{Sym}^{\lambda_k} V \longrightarrow V^{\otimes d - \lambda_k + 1} \otimes \mathrm{Sym}^{\lambda_{k-1}} V \big)
\end{aligned}$$

$$\begin{aligned}
\mathbb{S}^{\lambda/\tilde{\lambda}} V &= \mathrm{Im}\,\Big(\bigotimes_i \Lambda^{\mu_i - \tilde{\mu}_i} V \longrightarrow \otimes V^{\otimes d} \longrightarrow \bigotimes_j \mathrm{Sym}^{\lambda_j - \tilde{\lambda}_j} V \Big) \\
&= \mathrm{Im}\,\Big(\bigotimes_j \mathrm{Sym}^{\lambda_j - \tilde{\lambda}_j} V \longrightarrow \otimes V^{\otimes d} \longrightarrow \bigotimes_i \Lambda^{\mu_i - \tilde{\mu}_i} V \Big)
\end{aligned}$$